

HOANG BAO DUY LE

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EDUCATION

The University of Toledo, College of Engineering
Master of Science in Computer Science and Engineering

Expected Dec 2027

The University of Toledo, College of Engineering
Bachelor of Science in Computer Science and Engineering | GPA: 3.67

Dec 2022

PROFESSIONAL EXPERIENCE

System Software Developer | Nysus Solutions

Jan 2023 - Dec 2025

- Maintained and optimized Manufacturing Execution Systems (MES) and web-based manufacturing applications, integrating with MRP/ERP systems, PLCs, and HMIs to automate production and improve real-time data visibility.
- Collaborated with Application Architecture leadership to design, develop, and deploy scalable manufacturing software architectures supporting enterprise application growth and system reliability.
- Designed, and integrated MES solutions with MRP/ERP systems and shop-floor automation technologies (PLCs, HMIs, industrial equipment) to automate production processes and improve real-time data visibility.

System Software Developer Co-op | Nysus Solutions

Jan 2021 - Dec 2022

- Provided support and maintenance for enterprise manufacturing applications using Microsoft SQL Server Management Studio for troubleshooting, data analysis, and performance optimization.
- Configured and integrated PLC systems to automate industrial processes, ensuring seamless integration with existing manufacturing and customer production environments (DMS, DNUS, IAC).

CERTIFICATIONS

AWS Certified Cloud Practitioner — Amazon Web Services

PROJECTS

Multi-Agent Email Scam Detection System

Jan 2026 - Present

AI Engineer | Full Stack Developer

- Architected a multi-agent LLM system with a scam generation agent (GPT-4o mini), detection agent (Claude 3.5), and orchestration layer built using Microsoft Semantic Kernel.
- Designed agent communication protocols, structured messaging schemas, and shared memory workflows for reliable multi-step reasoning, agent coordination, and scalable AI task execution.
- Collaborate cross-functionally with architecture, backend, and frontend engineering teams to deliver scalable, maintainable, and extensible LLM-based application workflows and AI system integrations.

Multi-Agent Enterprise Automated AI Support

Sep 2025 - Present

Full Stack Developer

- Architected and deployed a multi-agent AI framework with specialized agents (Planning, Scanner, MCP, Specialist) operating in parallel to improve response accuracy and reasoning performance.
- Implemented dual-vector database architecture with OpenAI embeddings for semantic search across 50,000+ historical support tickets at 90%+ similarity thresholds.
- Developed an MCP agent that converts natural language to secure SQL, with automated schema discovery, multi-database routing, and aggregated result synthesis.

TECHNICAL SKILLS

Programming Languages: Java, C#, C++, Assembly Language, JavaScript, Python

Web Development: HTML5/CSS, PHP, React, NodeJS, MongoDB, Angular, MSSQL, ASP.NET

Cloud, AI & Tools: AWS (Cloud Practitioner Certified), Machine Learning, LLM, MCP, Git, SPSS, SAS, R, Blockchain